Nils Demougeot

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Seeking a Data Science / Machine Learning internship | Available April – August 2027 (4–5 months)

EDUCATION

ESILV – Engineering School Leonard de Vinci — Paris, France
Master of Engineering (Diplome d'Ingenieur), Data Science & Artificial Intelligence — *Sep 2023 – Jun 2028*
- Relevant coursework: Machine Learning, Neural Networks & Deep Learning (TensorFlow), Mathematics for Deep Learning, AI Algorithms, Statistics, Programming for Data Science

Riga Technical University — Riga, Latvia
Exchange Semester – Telecommunications & Data Transmission — *Aug 2025 – Jan 2026*

International School of Boston — Cambridge, MA
French Baccalaureate, Mathematics & Physics-Chemistry – Highest Honors — *Nov 2021 – Jun 2023*

PROFESSIONAL EXPERIENCE

Transdev – Cybersecurity & Data Analysis Intern — Paris, France
Cybersecurity team, Transdev France — *Jun 2025 – Aug 2025*
- Analyzed 200+ GB/day of file-transfer logs (CrowdStrike Falcon EDR, Netskope proxy) to detect data leaks
- Built 3 CrowdStrike Next-Gen SIEM dashboards and hourly alerts that surfaced previously undetected leaks
- Cut investigation time by ∼10+ hours each week; dashboards and alerts are still in production today
- Developed in-house tooling that avoided tens of thousands of euros in third-party CrowdStrike add-on costs

Massachusetts Institute of Technology (MIT) – Research Intern — Cambridge, MA
Research Laboratory of Electronics – Network Coding & Reliable Communications Group — *Jul 2022 – Aug 2022*
- Implemented Random Linear Network Coding in Python to model network data and optimize transmission protocols
- Simulated 40,000 packet generations (throughput, delay vs. loss rate), validated against a binomial model

PROJECTS

IBM & Probabl ML Hackathon – Parkinson's Motor-Score Bias Correction – Python, LightGBM, XGBoost
- Ranked 21st of 294 teams (∼1,000 students) on Kaggle (RMSE 2.50 vs. a 16.48 baseline), predicting an unbiased "true" OFF motor-severity score for Parkinson's patients from noisy, biased clinical visit data
- 4-person team; used IBM's Bob AI coding agent to rapidly prototype and iterate across 20+ feature-engineering and model-ensembling experiments under the hackathon's time constraints

Soccer Match Outcome Prediction (French Ligue 1) – Python, scikit-learn, pandas
- Merged 8 datasets (∼390K rows), engineered rolling form features with no data leakage, time-based validation
- Logistic regression, random forest, gradient boosting: 52% accuracy vs. 44% baseline, final 3-feature model

La Fabrique – Textile Upcycling Student Startup – Python, Django, PostgreSQL
- Led technical development for a 6-person team, from user research and business model to the final pitch to a jury
- Designed, built and deployed a mobile-first web app: garment photo analysis, sewing patterns, impact tracking
- Won the Startup Spirit Award among ∼140 teams at ESILV's annual project showcase

LEADERSHIP

DeVinci Durable – Head of Communications & Board Advisor — Paris, France
Student sustainability club — *2025 – Present*
- Lead a 10-person team creating content and awareness campaigns
- Organized the opening of a 600 m^2 (6,500 sq ft) urban community garden and a 4-day eco-seminar in Marseille

SKILLS & INTERESTS

Student clubs: Proteus Group (defense tech & engineering), LeoFly (aerospace & rocketry)

Programming: Python, SQL, JavaScript, PHP, HTML/CSS, C++, C#, Git

Data & ML: pandas, NumPy, scikit-learn, LightGBM, XGBoost, Matplotlib, seaborn, statistics & probability, CrowdStrike Next-Gen SIEM

Languages: French (native), English (fluent), Spanish (intermediate)